Worksheet 2.3 — Motion in One Dimension

Name: _____ Date: _____ Score: _____ / 30

Show all your work. Use $g = 9.8\ \text{m/s}^2$ and ignore air resistance. Round final answers to two decimal places where needed.

1. A car starts from rest and accelerates uniformly at **2.5 m/s²** along a straight road. (a) How long does it take to reach a speed of **20 m/s**? (b) How far does it travel in that time? (10 pts)

2. A stone is dropped from the top of a building **45 m** tall. (a) How long does it take to hit the ground? (b) What is its speed just before impact? (10 pts)

3. Two cyclists start **600 m** apart and ride toward each other along a straight path. Cyclist A rides at **5 m/s** and cyclist B rides at **7 m/s**. When and where do they meet? Measure the position from A's starting point. (10 pts)